Two Views of Topic Structure in Canadian Parliamentary Speech, 1920–1949

Draft for course seminar

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Abstract

We describe the topical content of 162,000 House of Commons speeches from 1920 to 1949 using two complementary methods: unsupervised KMeans on TF–IDF features ($k = 20$) and a supervised single-label classification produced by Gemini 2.0 Flash-Lite into ten policy categories. The unsupervised view recovers seven substantively interpretable clusters (taxation, railways, civil service, pensions, the wheat trade, banking and credit, and a Coal/Nova Scotia regional cluster); the remaining thirteen clusters are procedural or low-content residuals. The supervised view distributes most speech mass across *Other* (roughly 30–40% throughout), *Infrastructure*, *Trade*, and during 1940–1945 a sharp rise in *Wartime* that peaks near 50% of speeches at the height of the war. The two views agree on the timing and direction of the largest temporal feature in the period (the wartime spike) but disagree on its magnitude in a way the categorization granularity explains. We use the comparison to set up the labeled validation step required by the project's temporal extrapolation question.

1 Introduction

Canadian House of Commons debates between 1920 and 1949 span two world wars, the Great Depression, and the construction of the modern welfare state. The substantive topics that occupied Parliament must have moved over this period, but the magnitude and timing of those movements have not been documented at the speech level using a reproducible pipeline. This report asks a narrow descriptive question, in two forms: when the 162,000 MP-attributed speeches in our corpus are clustered into twenty unsupervised topics, do the clusters resolve to recognizable substantive themes and move plausibly over time; and when the same speeches are passed to a large language model and assigned to one of ten policy categories, what does the resulting time-series of category shares look like?

The answers turn out to be mutually informative. The unsupervised view recovers seven clusters with substantively legible top-term lists (taxation, railways, civil service, old-age pensions, the wheat trade, banking and credit, and a regional coal/Nova Scotia cluster) and one cluster with a sharp 1940–1945 spike consistent with wartime production debates. The remaining thirteen clusters are procedural or low-content residuals. The supervised view, in turn, distributes most speech mass across *Other*, *Infrastructure*, *Trade*, and *Budget*, with a striking near-50% peak in the *Wartime* category in 1942 followed by a tapering through 1945. Both views agree on the timing of the wartime spike. They disagree on its apparent magnitude in a way that follows mechanically from the differing granularity of the two label sets (one of twenty vs. one of ten).

This baseline matters for the temporal extrapolation exercise described in `docs/seminar_paper_guide.md`. Before asking whether a model trained on 1920–1949 predicts topics in 1950–1969, the topic structure of the training period itself needs to be characterized in both an unsupervised and a label-anchored frame. The three figures in this report are the simplest description the pipeline produces.

2 Data

The corpus is `output/speeches_all_mps.csv`, the merged speech-by-MP panel produced by the project pipeline. It contains 161,634 speech rows from MP-attributed sittings of the Canadian House of Commons covering the years 1920 to 1949. Each row carries the cleaned speech text, the date, the year and decade, and the speaker's party, province, gender, and riding from the linked ParlInfo MP biography.

Text preprocessing for the unsupervised view follows `src/prepare_claude_v2.py`: regex tokenization, English stopword removal, Porter stemming, and dropping tokens shorter than three characters. No part-of-speech filter is applied; the downstream TF–IDF vectorizer is left to weight informative terms.

For the supervised view, the cleaning is reversed: raw (unstemmed) speech text is recovered from the original `input/Speeches*.xlsx` files via the `basepk` key, alongside the ParlInfo `maintopic`, `subtopic`, and `subsubtopic` hints. Each speech is then passed to Gemini 2.0 Flash-Lite (`gemini-flash-lite-latest`, temperature 0, structured output schema constraining the response to exactly one of ten enum values). The implementation is in `src/classify_speeches_gemini.py`; speeches longer than 6,000 characters (well above the 99th percentile) are truncated. At the time of writing, 43,131 of the 161,634 speeches have been classified; the supervised analysis in Section 4.2 uses this labeled subset.

Two points are worth flagging. First, the panel is restricted to MP-attributed speeches, which excludes Speaker interventions and other procedural turns; it is not a complete record of the Hansard but the subset suited for speaker-level textual analysis. Second, the stemmed cleaning retains function-tokens such as *mr* and *hous*, which is why those tokens reappear later in cluster top-term lists and partly account for the procedural residual on the unsupervised side.

The classification prompt. The Gemini system instruction is reproduced verbatim in Figure 1. Each speech is then presented in a user message containing the year, speaker, party, riding, province, ParlInfo topic hint, and the raw (truncated) speech text. The structured-output schema forces the model to return exactly one of ten category strings, which removes the need to parse free-form responses.

3 Method

Unsupervised features and clustering. A TF–IDF vectorizer over unigrams and bigrams of the stemmed tokens, with English stopwords removed and a minimum document frequency threshold. KMeans with $k = 20$ is fit on the full document-term matrix with a fixed random seed; each speech receives a hard cluster label. The choice of $k = 20$ is not optimized: it is a coarse setting chosen to produce enough clusters to separate the major policy domains while leaving residual clusters as a visible diagnostic for low-content vocabulary. Implementation: `src/3_Model_Speech.py`.

Supervised classification. The classifier is Gemini 2.0 Flash-Lite called through the `google-genai` async client with a concurrency of 80 in-flight requests, temperature 0, and a structured-output enum schema. Re-runs are resumable: the script appends to `output/speeches_classified.csv` after every 500 successful calls and skips any `basepk` already present, so partial runs can be continued. Failed

```
You are a research assistant classifying speeches from the Canadian House
of Commons, 1920-1949. Read the speech and assign exactly ONE category
from this list:

- Wartime: WWI aftermath/veterans/pensions, WWII conduct/conscription/
  munitions/war finance, early Cold War defence and demobilisation.
- Infrastructure: railways, roads, ports, postal service, telegraph,
  public works, hydro/power, broadcasting infrastructure.
- Health care: public health, quarantine, hospitals, military medical,
  pensions for disability/illness, drug policy.
- Education: schools, universities, research grants, vocational
  training, residential schools.
- Income and Tax: income tax, excise, sales tax, customs duties as
  REVENUE, corporate tax, tax exemptions. (Tariffs as TRADE POLICY go
  under Trade.)
- International relations: treaties, League of Nations, UN, Commonwealth,
  diplomacy, embassies, foreign policy not centred on war.
- Budget: annual budget, supply, estimates, public debt, deficit,
  government spending overall. (Specific tax measures go under Income
  and Tax.)
- Trade: tariffs as trade policy, imperial preference, trade agreements,
  marketing boards, Wheat Board, export/import regulation.
- Immigration: immigration policy, refugees, displaced persons,
  citizenship, deportation, naturalisation.
- Other: anything that does not clearly fit the above (procedural
  motions, agriculture-not-trade, labour relations, indigenous affairs
  outside residential schools, constitutional questions, etc.).

Pick the SINGLE best fit. Use "Other" only if no listed category is a
reasonable match. Ignore procedural openings ("Mr. Speaker, ...") and
classify by the substantive topic the speaker addresses. When a speech
spans several categories, pick the one with the most content.
```

Figure 1: Gemini system instruction, verbatim from `src/classify_speeches_gemini.py`. The structured-output schema then constrains the response to exactly one of these ten category strings.

calls are retried up to six times with rate-limit-aware backoff; persistent failures are written to the output file with a non-empty `error` field. Implementation: `src/classify_speeches_gemini.py`.

Diagnostics. The unsupervised pipeline produces two artifacts: for each cluster, the ten terms with the largest centroid weight form a representative-words list (Figure 2); for each year-by-cluster cell, the share of that year's speeches assigned to that cluster is plotted as a time series (Figure 3). The supervised pipeline produces a third: for each year-by-category cell, the share of that year's labeled speeches in that category is plotted as a stacked area chart (Figure 4).

Two limits of the methods are load-bearing for the interpretation. TF–IDF + KMeans clusters by vocabulary, not by topic in any semantic sense: a cluster with high weight on *mr*, *hous*, and *speaker* is a cluster of procedural turns, not a topic cluster. The supervised pipeline, in turn, treats Gemini's single-label output as a measurement, not as ground truth; the ten categories are an editorial choice (made by the prompt author) and the model is free to fall back on *Other* when no fit is good. We treat agreement between the two views as evidence for a substantive pattern and

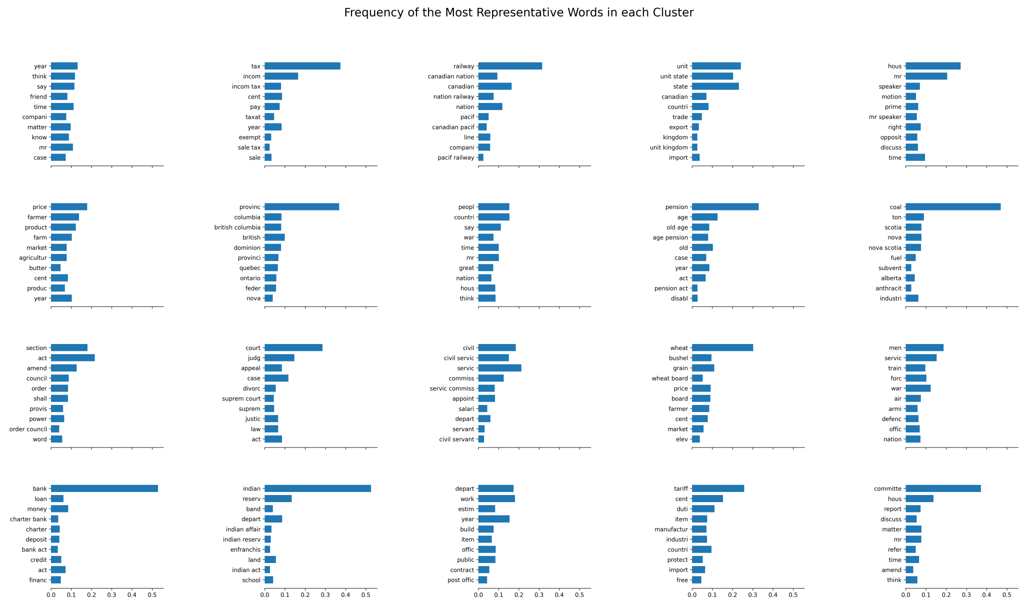

Figure 2: Top ten representative words for each of the twenty KMeans clusters, by centroid weight. Words are shown in stemmed form. Roughly a third of clusters have top terms that read as substantive policy topics; the rest are residual or procedural.

disagreement as a question to investigate, not as proof that one method is correct.

4 Results

4.1 Unsupervised cluster shapes

Figure 2 shows the top ten terms for each cluster. Six clusters resolve to clear substantive themes, summarized in Table 1: taxation, railways, civil service, old-age pensions, the wheat trade, and banking and credit. A seventh, dominated by *coal* and *nova scotia* vocabulary, picks out a regionally specific energy and resource debate that is narrower than the six above but still substantively interpretable.

Table 1: Seven KMeans clusters with substantively interpretable top-term lists, out of twenty total. Top terms are stemmed; the labels are assigned by the author on inspection of the top-ten terms per cluster (not generated by the model).

Label	**Top stemmed terms (selected)**
Taxation	tax, incom, incom tax, cent, pay, exempt, sale tax
Railways	canadian, canadian nation, nation railway, pacif, canadian pacif, line
Civil service	civil, civil servic, servic commiss, salari, civil servant, appoint
Old-age pensions	pension, old age, age pension, pension act, veteran
Wheat trade	wheat, bushel, grain, wheat board, elev, crop
Banking and credit	bank, loan, branch bank, charter bank, deposit, bank act
Coal and Nova Scotia	coal, ton, scotia, nova scotia, fuel, alberta

The remaining thirteen clusters fall into two groups. One group is procedural: top terms include

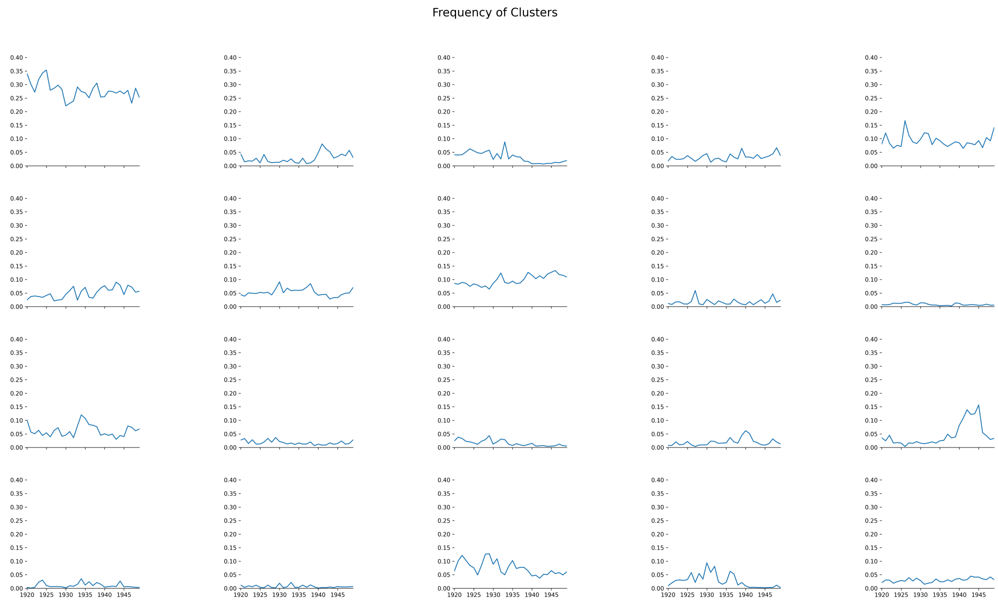

Figure 3: Share of each cluster in total speeches by year, 1920–1949. Each panel is one cluster on its own vertical scale (0 to 0.40). One cluster shows a sharp early-1940s spike consistent with wartime production debates; several others show smooth trends consistent with policy attention building or fading over the period.

mr, hous, motion, mr speaker, prime, oppos, discuss, capturing forms of debate rather than topics. These clusters are the residual that a stricter preprocessing pipeline would shrink. A second group has low and dispersed top-term weights, suggesting the cluster centroid sits in a low-density region of the term space. These are weak clusters in the sense that $k = 20$ is producing more groups than the vocabulary will support.

Figure 3 shows year-by-year shares. The most legible temporal pattern is the spike in the lower-right region of the grid, a cluster whose top terms point to wartime production: its share moves from near zero in the early 1930s to a peak above 0.15 in the early 1940s before falling back. This is the cleanest unsupervised match to a known historical shock in the data. Several other clusters show smoother trends. The pensions cluster grows through the late 1920s, consistent with the timing of the 1927 Old Age Pensions Act and its successors. The taxation cluster is high throughout the period but peaks in the mid-1940s, consistent with wartime revenue measures. A subset of clusters have essentially flat shares around 0.05 over the whole period; these are the procedural and residual clusters identified above.

The headline number from the unsupervised exercise is the resolution rate. Six of twenty clusters (30%) produce top-term lists that a domain reader can map to a recognizable policy topic without hesitation. Adding the regional-energy cluster gives seven of twenty (35%). The remaining thirteen clusters carry less than half of the total speech mass and consist primarily of procedural or low-content vocabulary.

4.2 Gemini category shares

Figure 4 shows the year-by-year share of each Gemini category. Three features stand out.

First, the *Wartime* share rises from below 5% before 1939 to roughly 45–50% in 1942, holds

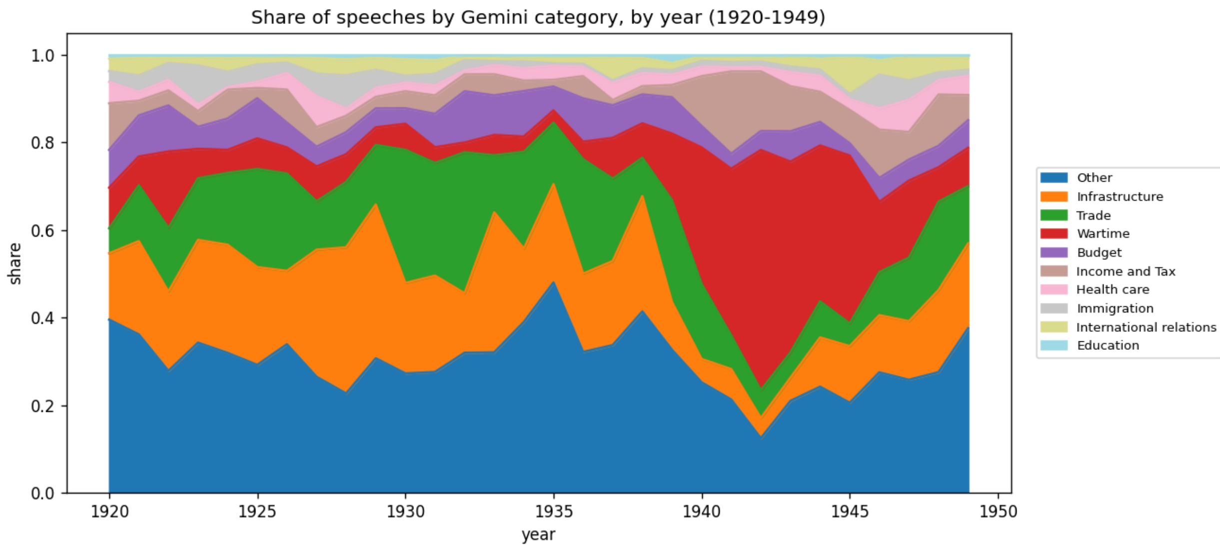

Figure 4: Share of speeches in each Gemini category by year, 1920–1949, stacked to 1.0. Computed on the 43,131-speech labeled subset. *Other* (blue, bottom) accounts for roughly 30–40% of speeches in most years. *Wartime* (red) rises sharply from 1940 to a peak around 1942, where it accounts for close to half of all classified speeches, before tapering through 1945.

high through 1944, and falls back below 20% by 1949. This is the dominant temporal feature in the labeled view of the period. The timing matches what a historian would expect: the September 1939 declaration of war is visible as the start of the rise, and the post-1945 decline (rather than a sharp drop) is consistent with the continued legislative aftermath of demobilization, veterans' benefits, and reconstruction.

Second, the *Other* category absorbs a striking 30–40% of the speech mass in most years. This is the natural home for procedural motions, indigenous affairs, labour relations, agricultural questions outside the wheat trade, and constitutional debates that the prompt explicitly routes to *Other*. Its size is therefore partly an artifact of where the ten-category schema draws its boundaries; it is not a model failure as such, but it sets an upper bound on how cleanly any downstream task can use the labeled subset.

Third, several substantively informative shifts are visible in the smaller categories. *Trade* falls during the wartime peak (roughly 1940–1944), consistent with wartime regulation displacing trade-policy debates from the floor; it recovers after 1945. *Infrastructure* runs at 10–20% throughout the period with a mild downward tilt during the war. *Income and Tax* and *Budget* together occupy a relatively stable share of around 10–15%. *Health care*, *Immigration*, *International relations*, and *Education* are individually small but non-trivial; the post-1945 rise in *International relations* is visible if narrow, consistent with the late-1940s diplomatic agenda.

4.3 Where the two views agree and disagree

The cleanest agreement between the two views is on the wartime feature. In the unsupervised view (Figure 3, lower-right panel) one cluster jumps from near zero to roughly 0.15 of all speeches at peak; in the supervised view (Figure 4, red band) the *Wartime* category jumps from near zero to roughly 0.45–0.50 at peak. The two magnitudes are not directly comparable: the unsupervised cluster competes with nineteen others for speech mass, while the supervised category competes with nine. What is comparable is the timing and the shape, both of which line up: rise in 1940, peak in 1942, decline through 1945. The *Wartime* category in the supervised view is, in effect, the supervised analog of the wartime cluster in the unsupervised view, and the two share the same temporal signature.

The cleanest disagreement is structural. The thirteen residual clusters in the unsupervised view, which carry less than half of the speech mass and which we have characterized as procedural or low-content, do not map cleanly to any single Gemini category. The natural alignment hypothesis (residual clusters $\rightarrow$ *Other*) is consistent with the size of *Other* in Figure 4 but cannot be checked from the descriptive statistics alone. A proper alignment table, with cluster $\times$ category counts on the labeled subset, is the next concrete step.

5 Discussion

Four implications follow.

The unsupervised pipeline as currently configured is a partial topic detector. Roughly a third of clusters resolve to substantive themes, which is enough to read off the broad shape of legislative attention but not enough to support a sharp downstream task. A useful next iteration would tighten preprocessing (explicitly drop procedural vocabulary, raise the minimum document frequency, consider sentence embeddings rather than TF–IDF) before reaching for a larger k.

The supervised view rests on a single-label decision into ten categories drawn by the prompt author. The *Other* share is informative about that choice: 30–40% of speech mass is being told there is no good fit. Two responses are open. The first is to refine the schema (add categories for indigenous affairs, agriculture, and labour relations as separate buckets) and re-run the classification on the residual. The second is to keep the schema fixed and treat *Other* as a measurement object in its own right, asking which procedural and policy domains it covers and how its share moves over the period. Both responses are useful; the prompt author's choice depends on what the downstream task is.

Where the two views overlap, the agreement is reassuring. The 1940–1945 wartime spike is the largest temporal feature in the labeled data and the most visible feature of the unsupervised cluster shares. The two pipelines, with different inputs (stemmed vs. raw text), different feature representations (TF–IDF vs. LLM embeddings), and different label schemata, agree on its timing and shape. This is the strongest evidence that whatever else is happening, neither method is hallucinating the wartime shift.

On the temporal extrapolation question, the figures contain a warning. If a supervised classifier trained on 1920–1949 with Gemini labels were applied to 1950–1969, the *Wartime* category would predict a share that has collapsed in the test period. Any drop in out-of-period accuracy will then be partly mechanical (the mix of categories has changed) and partly substantive (the vocabulary inside each category may have drifted). Decomposing the two requires category-conditional accuracy in the test period, as the seminar paper guide flags. The descriptive results here do not settle that question. They motivate it, and they identify the two cells whose temporal stability matters most for the extrapolation: *Wartime* (whose share is bounded above by zero in much of the test period) and *Other* (whose share will absorb whatever the schema fails to cover).

The natural next step is a labeled validation of the unsupervised pipeline against the Gemini classifications on the 43,131-speech overlap. Three diagnostics follow cheaply. First, an alignment table between the twenty clusters and the ten categories shows where the unsupervised pipeline already agrees with the LLM; cells with high agreement let the cluster proxy substitute for the LLM label on the unlabeled remainder of the corpus. Second, the residual clusters can be examined for category content: a procedural cluster whose speeches Gemini routes mostly to *Other* is consistent with the proceduralism hypothesis; a procedural cluster Gemini routes to a substantive category is more interesting and worth reading by hand. Third, the year-by-year share of the high-agreement clusters can be compared against the year-by-year share of the corresponding Gemini categories on the labeled subset, giving a measurement-error check on the descriptive series in Figure 3.

Limitations. The corpus excludes Senate speeches and Speaker interventions. The unsupervised preprocessing retains stemmed procedural vocabulary that drives a non-trivial number of clusters. The choice of $k = 20$ is not validated. The supervised view rests on a single-label, single-model classification at temperature 0 with no human gold standard for comparison; the model and the prompt author are both potential sources of systematic error, and the *Other* bucket may absorb a non-trivial amount of misclassification on top of legitimate residual content. Only 43,131 of the 161,634 speeches in the corpus have Gemini labels at the time of writing; the supervised series in Figure 4 is computed on this subset and is not population-weighted across years. All conclusions are descriptive.